

Q. what are Support vector Machines (SVM) ?

→ SVM is a supervised machine learning algorithm that works on both classification and regression problem statements.

- For classification problem statements, it tries to differentiate data points of different classes by finding a hyperplane that maximizes margins between the classes in the training data.

Q. what are Support vectors in SVM?

→ Support vectors are those instances that are located on the margin itself. For SVMs, the decision boundary is entirely determined using only the support vectors.

- Any instance that is not a support vector (not on the margin boundaries) has no influence whatsoever; you could remove them or add more instances; or move them around: as long as they stay off the margin they won't affect the decision boundary.
- For computing the predictions, only the support vectors are involved, not the whole training set.

Q. Types of SVM:

Linear SVM: when the data point is perfectly linearly separable only then we can use linear SVM. For example in 2D Perfectly linearly

separable means that the data points can be classified into 2 classes by using a single straight line.

Non-Linear SVM: when the data is not linearly separable we can use Non-Linear SVM. For example in 2D when the data point can't be separated into two classes using a straight line. In such cases we use kernel trick to classify them.

Q. Logistic Regression vs SVM:

Logistic Regression:

Probabilistic Approach: Logistic regression predicts the probability that an input belongs to a specific class (e.g. 80% chance of being spam). It uses sigmoid function to map inputs between 0 and 1.

Linear Decision Boundary: It assumes that the data can be separated by a straight line (on a hyperplane in higher dimensions). If the data is not linearly separable, Logistic Regression may struggle.

Simple and Interpretable: It's easy to implement and interpret, making it good starting point for classification problem. The coefficients of the model can also tell how each features influences the outcome.

SVM:

Maximizes the margin: SVM focuses on finding the decision boundary that maximizes the margin.

Handles Non-Linear Data: SVM can handle non linear data using "kernel trick", which transforms the data into a higher dimensional space where it becomes easier to separate.

Robust to Overfitting: By focusing on the points closest to the boundary, SVM is less likely to overfit, especially in smaller datasets.

Requires Tuning: SVM requires careful tuning of parameters (like the choice of kernel and regularization) to achieve optimal performance which can be time consuming.

Note: In SVM we don't have to worry about other observations since the margin is made using the points which are closest to hyperplane (support vectors), whereas in logistic regression the classified is defined over all points.

